

Progressive Web of Things: Design Concerns

Frances Berriman and Alex Russell were the first to come up with the concept of ‘progressive Web apps’. ‘Service worker’, ‘manifest’, ‘Web push’, and ‘add to the home screen’ were the first components of the PWA technological stack. The PWA is accessed and installed on devices using Web browsers. On smartphones, the prompts for PWAs are safer and more engaging. PWAs are being extended to the Internet of Things through the progressive Web of things. This article outlines a few design concerns with respect to the progressive Web of things.



Native applications for mobile devices are costly and time consuming to develop. A progressive Web app (PWA) has a responsive design. PWAs are popular among enterprises because they are simple to create and deploy. A native app can be used to access a PWA based Web application. The PWA icon can be found on the device’s home screen. It starts loading immediately, regardless of the state of the network. The app’s offline functionality and behaviour can be defined at the design and development stage. You can run the app without

using a browser. TLS connections are used to secure these apps. Service workers and a Web app manifest are used in PWAs. The manifest will have a name, *short_name*, *start_url*, icon, and display.

The responsive design of a PWA must account for form factors, screen sizes and mobile compatibility. To target tap, the design must take into account the correct viewport specification. On a standard device, the typical response time is roughly 5 seconds. After the service worker is installed, the response time improves. It is possible for the response time to be less than 2 seconds.

“PWAs are the future and working together on standards will make this easier, faster and less expensive in the future.”

—Eric Erway, Magento product manager

Visual transitions, fluid animations and user experience must all be considered in responsive design. Offline state management and behaviour must be considered during the design process. The offline